



Evaluating the Impact of Immersive Technologies on
STEM Education in Sri Lanka's
Secondary and Tertiary Curriculum

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Feasibility Study

Physics Lead

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1. Executive Summary

This feasibility study evaluates the development and implementation of a specialized Augmented Reality (AR) learning infrastructure aimed at enhancing **Grade 7 Physics education** within the Sri Lankan school system. Although the national science curriculum is well-structured, the teaching of physics at the middle-school level heavily depends on static two-dimensional textbook diagrams and verbal explanations to represent abstract and dynamic physical phenomena. Concepts such as **force and motion, light, heat and temperature, electricity, and energy transformations** are often difficult for students to visualize, leading to rote memorization rather than meaningful conceptual understanding.

The primary objective of this component is to design and develop a **mobile-based AR application** that enables students to visualize invisible and dynamic physics processes by overlaying interactive 3D simulations directly onto government-issued Grade 7 science textbooks. The proposed system allows learners to explore physical phenomena such as motion paths, light rays, heat transfer, and electric flow through real-time AR interactions, promoting inquiry-based and experiential learning.

The technical solution utilizes a smartphone camera to scan printed textbook diagrams, activating corresponding 3D physics simulations through **Unity and AR Foundation's marker-based image recognition**. These simulations allow students to observe, rotate, and manipulate virtual models, transforming static textbook illustrations into dynamic learning experiences. The system is designed specifically for the Sri Lankan context, ensuring compatibility with **low- to mid-range Android devices** and minimal reliance on additional infrastructure, making it suitable for under-resourced classrooms.

The feasibility assessment confirms both the technical and operational viability of the proposed solution. Performance evaluations indicate that optimized 3D rendering and lightweight AR assets can deliver smooth, real-time visualization even on entry-level smartphones, while offline functionality supports uninterrupted classroom use. A review of existing educational technologies reveals a clear research and application gap in **locally aligned AR-based physics tools** tailored to the Sri Lankan Grade 7 syllabus.

The findings support the advancement of the Physics component as a key pillar of the broader STEM AR learning platform. By making abstract physical concepts visible and interactive, this application has the potential to significantly improve conceptual understanding, increase student engagement, and encourage teacher adoption of digital pedagogy. Overall, the project presents a scalable, cost-effective, and pedagogically impactful solution for modernizing physics education in Sri Lanka.

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2. Introduction

2.1 Background

Physics at the Grade 7 level forms the foundation for scientific understanding; however, many students in Sri Lanka struggle with physics concepts due to their abstract and dynamic nature. Topics such as **force and motion, light, heat and temperature, electricity, and forms of energy** are primarily taught using lecture-based explanations and static two-dimensional diagrams in government-issued textbooks. These traditional teaching methods are limited in their ability to represent invisible processes, continuous motion, or cause-and-effect relationships, resulting in superficial understanding and reliance on memorization rather than conceptual learning.

Augmented Reality (AR) offers a practical solution to this challenge by overlaying interactive three-dimensional simulations directly onto printed textbook diagrams. For instance, a static illustration of a moving object can be transformed into a dynamic AR simulation showing motion paths, forces, and changes in velocity, or a light diagram can visualize reflection and refraction in real time. This approach converts passive observation into active exploration, allowing students to better understand physical phenomena through visual interaction and experimentation.

2.2 Objectives

- **Visualize Physical Phenomena:** Develop interactive AR models to demonstrate concepts such as force and motion, light behavior, heat transfer, and electrical flow.
- **Enhance Conceptual Understanding:** Enable students to observe invisible or dynamic processes through real-time 3D simulations.
- **Improve Student Engagement:** Encourage inquiry-based learning by allowing learners to interact with physics models rather than only reading diagrams.
- **Curriculum Alignment:** Ensure all AR content strictly follows the Sri Lankan Grade 7 Science syllabus (Parts I and II).

2.3 Scope

- **Technical:** Evaluate the performance of Unity and AR Foundation for rendering physics-based 3D models on low- to mid-range Android devices.
- **Operational:** Ensure the application is easy to use by Grade 7 students and integrates seamlessly with existing science textbooks.
- **Educational:** Address key learning difficulties in physics by focusing on visualization gaps in topics such as motion, energy, light, and electricity within the local classroom context.

3. Problem Statement

STEM education in Sri Lanka, particularly in **middle-school Physics**, faces significant challenges in effectively teaching concepts that involve invisible processes, motion, and dynamic interactions. Current Grade 7 Science instruction relies heavily on **static, two-dimensional textbook diagrams and lecture-based explanations** to describe topics such as **force and motion, electricity, light, heat, and energy transformations**. While these methods provide theoretical information, they are insufficient for helping students visualize how physical phenomena evolve over time or interact in real-world scenarios.

Many fundamental physics concepts—such as the motion of objects under applied forces, the flow of electric current in circuits, the refraction of light through different media, or the transfer of heat—are inherently dynamic and cannot be accurately represented through fixed images. As a result, students struggle to connect textbook diagrams with real physical behavior. Grade 7 represents a crucial stage where learners are first introduced to these abstract scientific ideas, yet the lack of interactive visualization often leads to misconceptions, confusion, and overreliance on rote memorization for examinations rather than true conceptual understanding.

Furthermore, practical laboratory facilities in many Sri Lankan schools are limited or unavailable, reducing opportunities for hands-on experimentation. This absence of experiential learning weakens student engagement and discourages interest in science and advanced STEM pathways. Teachers are constrained to explaining complex processes verbally or through drawings, which cannot adequately demonstrate continuous motion, cause-and-effect relationships, or invisible forces.

There is therefore a clear need for an **interactive, low-cost learning system** that can make abstract physics concepts visible, dynamic, and manipulable within the classroom environment. Augmented Reality (AR) technology provides a practical solution by overlaying interactive 3D simulations directly onto existing Grade 7 science textbook diagrams. By scanning textbook pages using a smartphone, students can observe and interact with simulations of physical phenomena in real time. This AR-based approach transforms learning from passive observation into active exploration, helping students develop deeper conceptual understanding while remaining accessible within Sri Lanka's resource-constrained educational context.

4. Project Description

The proposed Augmented Reality (AR)–based educational system is designed to enhance **Grade 7 Physics learning** by integrating real-time 3D simulations into the existing Sri Lankan science textbook framework. As the **Physics Lead**, this component focuses on visualizing physical phenomena that are difficult or impossible to observe directly through traditional classroom instruction, such as motion, force interactions, electrical flow, light behavior, and energy transfer. By transforming static textbook diagrams into interactive AR experiences, the system enables students to explore physics concepts dynamically rather than relying solely on theoretical explanations.

A core feature of the system is its use of **marker-based AR**, where textbook images act as triggers for launching interactive 3D simulations via mobile devices. Using Unity and image-target recognition, students can scan diagrams related to topics such as **force and motion, static electricity, simple electrical circuits, refraction of light, heat transfer, and sound waves**. Once activated, these simulations allow learners to observe cause-and-effect relationships in real time, manipulate parameters (such as force direction, angle of motion, circuit components, or light paths), and view processes from multiple perspectives to deepen conceptual understanding.

The Physics AR module is carefully aligned with the **Sri Lankan Grade 7 Science syllabus (Part I and Part II textbooks)** to ensure curricular relevance and classroom applicability. Guided AR learning sequences provide step-by-step conceptual explanations synchronized with visual simulations, while interactive feedback encourages experimentation and inquiry-based learning. This approach supports both teacher-led instruction and self-paced student exploration.

By offering a **low-cost, mobile-based solution** optimized for mid-range Android devices, the system is well suited for resource-constrained school environments where laboratory access is limited. Overall, this Physics AR component aims to shift learning from passive observation to active exploration, reduce common misconceptions, increase student engagement, and support teachers in delivering more effective and meaningful physics instruction within Sri Lankan classrooms.

5. Feasibility Analysis

5.1 Technical Feasibility

The technical analysis confirms that the proposed **Grade 7 Physics AR application** is feasible using currently available development tools and commonly used mobile devices. The system will be developed using **Unity** as the core development platform, **Vuforia** for marker-based image recognition of textbook diagrams, and **Blender** for creating accurate 3D physics models such as force vectors, electrical circuits, light rays, and motion paths. These tools are industry-standard, well-documented, and suitable for educational AR development.

From a performance perspective, the application is designed to operate efficiently on **mid-range Android smartphones**, which are widely used by Sri Lankan students. The AR models and animations will be optimized to ensure smooth rendering, stable tracking, and low memory usage. Special attention will be given to maintaining **scientific accuracy**, as physics simulations must correctly represent real-world behaviors such as motion, energy transfer, refraction, and electrical flow to preserve educational credibility.

5.2 Market Feasibility

The market feasibility analysis indicates a strong demand for **innovative physics teaching tools** in Sri Lankan schools, particularly at the **Grade 7 level**, where students are first introduced to fundamental physics concepts. Current teaching methods rely heavily on static textbook diagrams and verbal explanations, which are insufficient for explaining dynamic phenomena such as force and motion or light behavior.

There is a clear **research and implementation gap** in localized AR solutions that align directly with the **Sri Lankan Grade 7 science syllabus**, especially in the Sinhala-medium context. Existing international AR tools are either not curriculum-aligned or require advanced hardware. This creates a strong opportunity for a low-cost, syllabus-based Physics AR solution targeted at Grade 7 students and science teachers.

5.3 Financial Feasibility

The project is financially feasible, as the estimated development cost ranges between **LKR 20,000–30,000**, primarily covering minimal software-related expenses and testing resources. Most development tools used, such as Unity and Blender, offer free or educational licenses, significantly reducing costs.

Since the application is designed to function **offline after installation**, it eliminates recurring internet-related expenses. Additionally, the use of students' existing smartphones avoids the need for schools to invest in expensive laboratory equipment or specialized AR hardware, making the solution economically practical for both urban and rural schools.

5.4 Operational Feasibility

Operationally, the system can be implemented with a **small multidisciplinary student research team**, including AR developers, physics content designers, and basic UI/UX contributors. The application is designed to integrate seamlessly with **existing Grade 7 science textbooks**, allowing teachers to continue their usual teaching practices while enhancing lessons with AR support.

The app is simple to use and does not require advanced technical skills from teachers or students. Basic orientation sessions will be sufficient to familiarize users with scanning textbook images and interacting with simulations. Since the system runs on standard smartphones and works without internet connectivity, it is suitable for use in both **urban and rural school environments**. Continuous improvements can be made using feedback from teachers and students to ensure long-term usability and relevance.

5.5 Environmental and Ethical Feasibility

From an environmental perspective, the AR-based learning system has a **minimal ecological footprint**. It reduces dependence on physical laboratory materials and printed resources, thereby limiting waste and resource consumption. Digital simulations replace repetitive physical experiments, promoting a more sustainable approach to science education.

Ethically, all physics content is developed and reviewed in alignment with the **Sri Lankan national curriculum** to ensure accuracy and appropriateness for Grade 7 learners. The application does not collect personal student data, ensuring **data privacy and security**. The system is designed to be inclusive and accessible, supporting equal learning opportunities for students regardless of location or school resources.

06. Risk Analysis

6.1 Technical Risks

One of the primary technical risks is **device compatibility**. The AR application must function smoothly across a wide range of Android smartphones, including older and low-end models commonly used by Sri Lankan students. Variations in camera quality, processing power, and sensor accuracy may affect AR tracking performance if not properly optimized.

Another key risk is **software stability and performance**. The application must maintain smooth, real-time 3D rendering of physics simulations such as force vectors, motion paths, light refraction, and electrical circuits. Poor optimization could lead to lag, crashes, or inaccurate animations, which would reduce learning effectiveness.

Image recognition reliability is also critical. Since the system relies on Vuforia to recognize textbook diagrams, inconsistent lighting conditions, page damage, or low print quality may reduce detection accuracy. This could prevent AR models from triggering correctly during classroom use.

Finally, **scientific accuracy** presents a major risk. Physics simulations must correctly represent real-world laws and behaviors. Any inaccuracies in motion, energy transfer, or optical behavior could mislead students and reduce the educational credibility of the application.

6.2 Financial Risks

The project involves **initial development costs**, including time investment for AR development, physics-accurate 3D modeling, testing, and potential software-related expenses. Although the project is low-cost, limited funding may restrict the level of detail or the number of physics topics included.

Another financial risk is **adoption constraints**. Some schools may be hesitant to use AR-based learning tools due to budget limitations, lack of institutional support, or concerns about student access to smartphones. This could affect large-scale deployment despite the technical feasibility of the solution.

6.3 Operational Risks

A significant operational risk is **user training and acceptance**. Teachers and students who are unfamiliar with AR technology may require initial guidance to effectively integrate the application into physics lessons. Without proper orientation, the technology may be underutilized.

Maintenance and updates also pose a risk. The application will require periodic updates to remain compatible with newer Android versions and device hardware. Limited technical support resources could affect long-term sustainability.

Lastly, **scalability challenges** exist if the system is expanded to include additional Grade 7 science lessons or higher-grade physics topics. Scaling the platform would require additional development effort, physics content validation, and testing, which may strain available resources.

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